

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2076

Bioenergy Power Llc 138 kV
March 29, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2076. The Interconnection Customer proposes a 435.4 MW battery energy storage facility interconnecting at the Bioenergy Power Llc 138 kV substation in the Pennsylvania area. The request was received on May 02, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 503 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 35 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$121,354,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Big Horn – Johanna Substation 138.0 kV
Contingency: Casa Blanca Substation – Thorn 138.0 kV
Loading: 211.5% (Rating: 187 MVA)
DFAX: 65.1%
Pre-project loading: 59.8%
2. Facility: Bioenergy Power Llc – Johanna 138.0 kV Line
Contingency: Casa Blanca – Thorn 138.0 kV
Loading: 182.8% (Rating: 246 MVA)
DFAX: 100.0%
Pre-project loading: 5.5%
3. Facility: Casa Blanca – Thorn 138.0 kV
Contingency: Johanna – Bioenergy Power Llc 138.0 kV
Loading: 180.6% (Rating: 249 MVA)
DFAX: 100.0%
Pre-project loading: 5.5%
4. Facility: Casa Blanca – Bioenergy Power Llc 138.0 kV
Contingency: Bioenergy Power Llc – Johanna 138.0 kV line
Loading: 179.1% (Rating: 247 MVA)
DFAX: 100.0%
Pre-project loading: 2.8%
5. Facility: Honeygo – Norway In 138.0 kV Line
Contingency: Newberry 345/138kV Transformer
Loading: 158.8% (Rating: 207 MVA)
DFAX: 29.0%
Pre-project loading: 98.0%

6. Facility: Roadrunner – Honeygo 138.0 kV
Contingency: Newberry 345/138kV Auto
Loading: 136.8% (Rating: 211 MVA)
DFAX: 29.0%
Pre-project loading: 77.0%
7. Facility: Neoga – Ramos 138.0 kV
Contingency: Demoss – Neoga 138.0 kV
Loading: 130.5% (Rating: 251 MVA)
DFAX: 7.6%
Pre-project loading: 117.4%
8. Facility: Big Horn – West Pt 138.0 kV
Contingency: Casa Blanca – Thorn 138.0 kV Line
Loading: 130.1% (Rating: 337 MVA)
DFAX: 36.9%
Pre-project loading: 82.4%
9. Facility: Shawan Rd – Johanna 138.0 kV
Contingency: Big Horn – Brinkerhoff 138.0 kV
Loading: 123.5% (Rating: 243 MVA)
DFAX: 23.3%
Pre-project loading: 81.8%
10. Facility: Caney Crk – Bge Timonium Substation 138.0 kV
Contingency: Cipres – Thompson Peak Tap 138.0 kV L
Loading: 122.1% (Rating: 264 MVA)
DFAX: 13.3%
Pre-project loading: 100.2%
11. Facility: Ramos – Illinois 4 138.0 kV
Contingency: Neoga – Cipres 138.0 kV
Loading: 121.4% (Rating: 231 MVA)
DFAX: 8.2%
Pre-project loading: 105.9%
12. Facility: Demoss – Neoga 138.0 kV
Contingency: Ramos – Neoga Substation 138.0 kV
Loading: 119.6% (Rating: 294 MVA)
DFAX: 7.8%
Pre-project loading: 108.1%
13. Facility: Texas – Shawan Rd 138.0 kV
Contingency: Brinkerhoff – Big Horn 138.0 kV
Loading: 117.9% (Rating: 229 MVA)
DFAX: 23.3%
Pre-project loading: 73.7%
14. Facility: Tenney – Demoss 138.0 kV #2
Contingency: Demoss – Tenney 138.0 kV Cir. 1
Loading: 117.9% (Rating: 324 MVA)
DFAX: 6.9%
Pre-project loading: 108.6%
15. Facility: Brinkerhoff Substation – Big Horn 138.0 kV
Contingency: Johanna – Shawan Rd 138.0 kV
Loading: 116.0% (Rating: 288 MVA)
DFAX: 22.8%
Pre-project loading: 81.4%
16. Facility: Demoss – Ramos 138.0 kV Line 2
Contingency: Ramos – Demoss 138.0 kV #1

Loading: 115.5% (Rating: 354 MVA)

DFAX: 7.2%

Pre-project loading: 106.6%

17. Facility: West Pt – Powell 138.0 kV

Contingency: Thorn Substation – Casa Blanca 138.0 kV

Loading: 114.7% (Rating: 363 MVA)

DFAX: 25.0%

Pre-project loading: 84.8%

18. Facility: Neoga – Cipres 138.0 kV

Contingency: Hidalgo Smelter 345 kV/1 kV Transformer

Loading: 114.1% (Rating: 445 MVA)

DFAX: 10.7%

Pre-project loading: 103.7%

19. Facility: Texas – Mays Chapel 138.0 kV

Contingency: Brinkerhoff – Big Horn 138.0 kV line

Loading: 113.9% (Rating: 224 MVA)

DFAX: 23.3%

Pre-project loading: 68.6%

20. Facility: Roadrunner – Thorn 138.0 kV Line

Contingency: Newberry 345-138 kV Auto

Loading: 113.2% (Rating: 257 MVA)

DFAX: 39.4%

Pre-project loading: 46.7%

21. Facility: Illinois 4 – Nashville 138.0 kV

Contingency: Cipres – Neoga 138.0 kV L

Loading: 112.7% (Rating: 217 MVA)

DFAX: 8.2%

Pre-project loading: 96.2%

22. Facility: Iatan – Cipres 138.0 kV L

Contingency: Cipres – Neoga 138.0 kV

Loading: 111.8% (Rating: 181 MVA)

DFAX: 8.7%

Pre-project loading: 91.1%

23. Facility: Rustic – Tenney 138.0 kV

Contingency: Tnney – Rustic 138.0 kV

Loading: 111.7% (Rating: 344 MVA)

DFAX: 6.7%

Pre-project loading: 103.2%

24. Facility: Tenney – Rustic 138.0 kV

Contingency: Rustic – Tenney 138.0 kV Line 2

Loading: 111.5% (Rating: 344 MVA)

DFAX: 6.7%

Pre-project loading: 103.0%

25. Facility: Demoss – Tenney 138.0 kV L

Contingency: Tenney – Demoss 138.0 kV Line 2

Loading: 110.2% (Rating: 346 MVA)

DFAX: 6.9%

Pre-project loading: 101.5%

26. Facility: Westwood Generation Llc 500/1 kV Transformer

Contingency: Foster Wheeler Mt Carmel Cogen 500 kV/345 kV XFMR

Loading: 110.0% (Rating: 1,124 MVA)

- DFAX: 5.8%
Pre-project loading: 107.8%
27. Facility: Ramos – Demoss 138.0 kV Cir. 1
Contingency: Ramos – Demoss 138.0 kV Cir. 2
Loading: 108.9% (Rating: 375 MVA)
DFAX: 7.2%
Pre-project loading: 100.6%
28. Facility: Fort Dodge – Hygiene 138.0 kV L
Contingency: Bioenergy Power Llc – Johanna 138.0 kV
Loading: 107.8% (Rating: 226 MVA)
DFAX: 13.0%
Pre-project loading: 82.6%
29. Facility: Demoss – Tenney 138.0 kV line
Contingency: Demoss – Tnney 138.0 kV
Loading: 107.1% (Rating: 356 MVA)
DFAX: 6.9%
Pre-project loading: 98.7%
30. Facility: Westwood Generation Llc 345 kV/1 kV Transformer Bank
Contingency: Foster Wheeler Mt Carmel Cogen Substation 500-345 kV Autotransformer
Loading: 105.8% (Rating: 1,170 MVA)
DFAX: 5.8%
Pre-project loading: 103.6%
31. Facility: Nashville – Lions Bay 138.0 kV
Contingency: Cipres – Neoga Substation 138.0 kV line
Loading: 105.6% (Rating: 194 MVA)
DFAX: 8.2%
Pre-project loading: 87.2%
32. Facility: Zachary – Levee 138.0 kV
Contingency: Talenenergy Martins Crk Llc Harrisburg 500/345 kV XFMR
Loading: 105.1% (Rating: 379 MVA)
DFAX: 7.1%
Pre-project loading: 96.9%
33. Facility: Mays Chapel – Iatan 138.0 kV L
Contingency: Big Horn – Brnkerhoff 138.0 kV L
Loading: 105.0% (Rating: 211 MVA)
DFAX: 23.3%
Pre-project loading: 57.0%
34. Facility: Bge Timonium – Brinkerhoff 138.0 kV
Contingency: Johanna – Shawan Rd 138.0 kV
Loading: 104.2% (Rating: 284 MVA)
DFAX: 22.8%
Pre-project loading: 69.1%
35. Facility: For Dodge – Deer Trail 138.0 kV
Contingency: Johanna – Bioenergy Power Llc 138.0 kV
Loading: 100.5% (Rating: 287 MVA)
DFAX: 10.2%
Pre-project loading: 85.1%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Big Horn – Johanna 138.0 kV line	Rebuild line with higher-capacity conductor	\$2,822,000	99% (\$2,795,000)
Bioenergy Power Llc – Johanna 138.0 kV Line	Rebuild line with higher-capacity conductor	\$23,428,000	100% (\$23,428,000)
Casa Blanca – Thorn 138.0 kV	Rebuild line with higher-capacity conductor	\$2,681,000	100% (\$2,681,000)
Bioenergy Power Llc – Casa Blanca 138.0 kV	Rebuild line with higher-capacity conductor	\$3,798,000	100% (\$3,798,000)
Norway In – Honeygo 138.0 kV	Rebuild line with higher-capacity conductor	\$1,702,000	100% (\$1,702,000)
Honeygo Substation – Roadrunner Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$6,090,000	100% (\$6,090,000)
Neoga – Ramos 138.0 kV	Rebuild line with higher-capacity conductor	\$1,697,000	76% (\$1,297,000)
Big Horn – West Pt 138.0 kV	Rebuild line with higher-capacity conductor	\$7,330,000	94% (\$6,857,000)
Johanna – Shawan Rd 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,097,000	93% (\$1,025,000)
Caney Crk – Bge Timonium Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,208,000	66% (\$1,453,000)
Ramos – Illinois 4 138.0 kV	Rebuild line with higher-capacity conductor	\$6,133,000	92% (\$5,611,000)
Neoga – Demoss 138.0 kV	Rebuild line with higher-capacity conductor	\$3,356,000	77% (\$2,584,000)
Texas – Shawan Rd Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$3,776,000	93% (\$3,526,000)
Demoss – Tenney 138.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$806,000	87% (\$700,000)
Brinkerhoff Substation – Big Horn 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,867,000	92% (\$1,710,000)
Demoss – Ramos 138.0 kV Ckt 2	Rebuild line with higher-capacity conductor	\$2,379,000	87% (\$2,068,000)
West Pt – Powell 138.0 kV	Rebuild line with higher-capacity conductor	\$14,760,000	74% (\$10,901,000)
Cipres Substation – Neoga 138.0 kV	Rebuild line with higher-capacity conductor	\$1,456,000	72% (\$1,054,000)
Texas – Mays Chapel 138.0 kV	Rebuild line with higher-capacity conductor	\$4,018,000	93% (\$3,753,000)
Thorn – Roadrunner 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,139,000	100% (\$5,139,000)
Nashville – Illinois 4 138.0 kV	Rebuild line with higher-capacity conductor	\$2,435,000	92% (\$2,227,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cipres – Iatan 138.0 kV	Rebuild line with higher-capacity conductor	\$1,086,000	78% (\$852,000)
Rustic – Tenney 138.0 kV	Rebuild line with higher-capacity conductor	\$4,079,000	87% (\$3,547,000)
Tenney – Rustic 138.0 kV	Rebuild line with higher-capacity conductor	\$4,821,000	87% (\$4,191,000)
Demoss – Tenney 138.0 kV L	Rebuild line with higher-capacity conductor	\$1,157,000	87% (\$1,006,000)
Westwood Generation LLC 500/1 kV Transformer	Replace transformer with higher-rated unit	\$28,794,000	19% (\$5,356,000)
Ramos – Demoss 138.0 kV	Reconductor line	\$729,000	87% (\$634,000)
Fort Dodge – Hygiene 138.0 kV L	Reconductor line	\$241,000	100% (\$241,000)
Demoss – Tenney 138.0 kV Cir. 3	Reconductor line	\$550,000	87% (\$478,000)
Westwood Generation LLC 345/1kV Auto	Replace transformer with higher-rated unit	\$8,412,000	19% (\$1,565,000)
Nashville – Lions Bay 138.0 kV	Reconductor line	\$697,000	92% (\$637,000)
Zachary – Levee 138.0 kV	Reconductor line	\$6,424,000	77% (\$4,959,000)
Iatan – Mays Chapel 138.0 kV	Reconductor line	\$464,000	93% (\$434,000)
Brinkerhoff – Bge Timonium 138.0 kV Line	Reconductor line	\$1,675,000	92% (\$1,534,000)
For Dodge – Deer Trail 138.0 kV	Reconductor line	\$6,195,000	89% (\$5,521,000)
	Total allocated to project		\$121,354,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.